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Engineering & Technology for Holistic Restructuring**
"New Vistas in Technological Advancements for Globalization"

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A New Digital Encryption and Decryption Process using Bit Shifting & 3*3 Matrix XOR Operation Methodology

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Abstract

In today's technological world the information security could be a difficult issue. There is a requirement for a stronger encoding that is extremely exhausting to crack. Moreover, in a Bit Shifting and Stuffing (BSS) system to represent a printable character it only needs 7 bits as per its ASCII value. In the computer system to represent a printable character it requires one byte, i.e. 8 bits. It is clear from the aforementioned expressions that a printable character occupies '7' bits and the last bit value is '0', which is not useful for the character. In BSS method it is stuffing a new bit in the place of unused bit, which is shifting from another printable character. Thus, in this BSS methodology after encryption, for every eight bytes of plain text it will generate seven bytes' cipher text and in decryption, for every seven bytes of cipher text, it will reproduce eight bytes of plain text. The present research proposed a new replacement algorithmic rule for Digital encoding called as 'Bit Shifting and Matrix XOR Operation Conversion Technique', which increases the complexity of encryption of the data as well as form of the data. The experimental results showed that the new theme is very fast in encoding and safer.